

# Clinical Pharmacogenomics (PGx) Safety Report

|                      |                                               |
|----------------------|-----------------------------------------------|
| Patient File:        | patient_sample1.vcf                           |
| Proposed Medication: | Codeine                                       |
| Analysis Type:       | Tri-Domain PGx & Organ Clearance Safety Check |

## Genomic & Guideline Findings

- **Gene Evaluated:** CYP2D6
- **Assessed Phenotype:** Poor Metabolizer
- **CPIC Recommendation:** Avoid Codeine due to lack of analgesic effect (analgesic failure).
- **FDA Note:** FDA Note: Reduced conversion to morphine; potential reduced efficacy.
  
- **Gene Evaluated:** CYP2C19
- **Assessed Phenotype:** Poor Metabolizer
- **CPIC Recommendation:** No specific CPIC rule found for this variant.
- **FDA Note:** No specific FDA table entry found for this variant.
  
- **Gene Evaluated:** SLCO1B1
- **Assessed Phenotype:** Poor Metabolizer
- **CPIC Recommendation:** No specific CPIC rule found for this variant.
- **FDA Note:** No specific FDA table entry found for this variant.

## Organ Clearance & Physiological Parameters

- **Kidney eGFR:** Normal (Value: 90 mL/min)
- **Liver ALT:** Normal (Value: 25 U/L)

**Anatomical Warnings:**

- Organ clearance parameters (Kidney & Liver) are within normal physiological ranges.

**FINAL CLINICAL VERDICT: SAFE / NORMAL**